HUY TRINH

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EDUCATION

University of California San Diego | *B.S, Data Science - GPA: 3.95* **August 2020 - June 2023**

EXPERIENCE

Workday | *Data Analytics Engineer* **September 2023 - Present**

- Enhanced efficiency for **200+ salespeople** with a RAG LLM sales chatbot with **FastAPI**, functional test, **Grafana**-like monitor dashboards, and improved scalability by migrating chat storage from **Databricks** to **MongoDB** & **DynamoDB**
- Automated reporting over **200,000** data errors and missing data points by developing **Snowpark Python** scripts on **Snowflake tasks (DAG)** & **dbt** to prevent downstream reporting failures
- Identified **$100k+** in annual **compute savings** by benchmarking AWS vs. Databricks LLM inference costs and steering infrastructure decisions to non-technical stakeholders
- Optimized data processing runtime by **80%** (**4 hours** to **~45 minutes**) and enhanced feature development by refactoring legacy Pandas codebase into reusable **PySpark** jobs and **SQL** queries on Databricks.
- Data modeled and deployed a prompt management system by using **SQLAlchemy** and **MongoDB** to version, track, and roll back LLM prompts, hence accelerating prompt tuning, and experimentation
- Reduced manual sales operations by automating a DLT workflow pipeline with LLM runtime calls, enabling efficient analysis and summarization of customer notes.
- Drove an **8% lift in conversion** by leading model features development, & deploying cross-sell & **churn** models with **H2O**, **MLflow** & **Databricks AutoML**, focusing on experiment tracking, model selection, & production ML pipelines
- Projected **+$200K** in next-year cost savings by leading the enterprise **migration** from **H2O.ai** to **Databricks AutoML** to streamline ML pipelines.

Workday | *Data Analyst Intern* **July 2022 - September 2022**

- Achieved **78%** accuracy in predicting fields for **internal incident ticketing** by developing a multiclass classification model with ServiceNow Predictive Intelligence, outperforming the benchmark model
- Revamped field predictions display by using JavaScript with Server-side Scripting GlideRecord API

Wells Fargo | *Data Analyst Intern* **June 2021 – August 2021**

- Engineered Tableau dashboards for service team's agile resource allocation, finding bottlenecks in time allocation
- Reduced customer behavior analysis query time by **25%**, optimizing **Teradata SQL** queries and developed **Tableau** dashboards for anomaly detection in bank transaction events.

PROJECTS

NYC Taxi Data Lakehouse

- Developed a hybrid batch-stream data pipeline for NYC Taxi trip data using **Kafka**, **Flink**, and **Debezium** to capture CDC changes from PostgreSQL, enabling real-time fare analytics and anomaly detection
- Architected a lakehouse solution with **Delta Lake**, **Trino**, and **MinIO**, and automated workflows with **Airflow** to support ingestion, transformation, and ML feature generation for demand forecasting

Decentralized Escrow Marketplace | associated with **Franklin Templeton |** Link

- A crypto e-commerce marketplace where users can connect crypto wallets, publish products, deposit ETH to buy, approve or reject purchase, track transactions and status, confirm receipt and rate products
- Implemented a responsive front-end software application using **React.js, Vite, JSX, Tailwind CSS, HTML** and backend contract using **Solidity** and **Thirdweb API**

TECHNICAL SKILLS

Languages: Python, SQL, Tableau, HTML, CSS, Javascript, d3.js, R, C++, Stata, Java

Library/Framework: Pyspark, MLflow, Node JS, React, Django, Sklearn, NLTK, Tensorflow, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Selenium

Tools: Databricks, Snowflake, MongoDB, AWS DynamoDB, AWS S3, Jupyter, Github, Terraform, Docker, Cassandra, Neo4j, Redis, dbt, New Relic, Kafka, gRPC, REST API, Grafana, H2O.ai

Certifications: Databricks Certified Data Engineer Associate, Snowflake badges